

ENHANCING RISK AWARENESS IN LLM AGENTS VIA PROBING SAFETY BOUNDARIES

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ABSTRACT

Large Language Models (LLMs) have evolved to function as autonomous agents capable of real-world tasks, including multi-step reasoning, tool use, and decision-making. While promising for complex task execution, such capabilities introduce unpredictable risks, highlighting the need for LLM agents to maintain internal risk awareness throughout task processes. We propose a judgment mechanism that integrates harmfulness and consistency metrics to assess agents' self-awareness of risk through multiple interactions. We then apply information-theoretic analysis to determine safety boundaries, which are thresholds that separate different behavioral patterns of LLM agents, and provide a natural means for organizing agent behaviors into consistent safety, uncertainty, and consistent unsafety without manual division or binary classification. Leveraging identified boundaries, we develop a dual-objective training strategy for safety alignment that employs supervised fine-tuning to strengthen consistent safety responses and unlearning techniques to reduce harmful behaviors. Experimental evaluation demonstrates that our method effectively reveals and improves internal risk awareness of LLM agents, with their responses moving toward more consistent and safer behaviors compared to existing approaches.

Index Terms— LLM agent, AI safety, Information theory

1. INTRODUCTION

Large Language Models (LLMs) have transformed artificial intelligence, extending far beyond basic language tasks such as text generation. By incorporating multi-step reasoning, tool use, and memory, LLMs have evolved into autonomous agents capable of perceiving, understanding, and interacting with their environment [1, 2, 3]. These capabilities enable them to undertake real-world tasks including code development, data analysis, and automated decision-making in dynamic scenarios. While such advancements make LLM-based agents highly promising for complex task execution, they introduce considerable safety risks and vulnerabilities [4]. Agents can be compromised by prompt injection attacks that manipulate their decision-making processes, causing them to generate harmful or inappropriate content that violates safety guidelines [5]. They may comply with malicious instructions hidden within seemingly benign requests, leak sensitive information from training data or interaction history, produce inconsistent or unreliable responses, or execute harmful actions via tool use [6, 7]. Moreover, the autonomous nature of these agents amplifies the potential consequences of such failures, as they can take real-world actions with significant impact [8, 9]. Therefore, the ability to accurately perceive and respond to harmful information is critical for reliable and responsible functioning. A central challenge is to understand how LLM agents recognize, interpret, and react to potentially unsafe

or risky tasks, an essential step toward maintaining consistent and trustworthy behaviors.

Existing frameworks like AgentHarm [10], InjecAgent [11], and TaskShield [12] achieve reasonable detection rates and offer basic protection. While these frameworks handle complex threats like indirect prompt injection and tool manipulation, they lack mechanisms to quantify agents' internal confidence or uncertainty in safety assessments. These methods provide only output-level judgments without capturing how agents internally reason about safety [13, 14]. Although recent work has progressed in uncertainty quantification for LLMs in general tasks [15, 16], current approaches lack mechanisms to assess agents' confidence in safety-related decisions specifically. While general uncertainty methods estimate model confidence in factual or reasoning tasks [17, 18], they do not address the unique challenges of safety awareness in agent interactions. Without measuring safety-specific uncertainty, agents cannot calibrate responses based on their certainty about potential risks, limiting both reliability and interpretability in safety-critical scenarios. This raises three fundamental questions: i) *Can LLM agents identify risks and provide safe responses?* ii) *Do LLM agents consistently generate safe responses when facing similar harmful scenarios?* iii) *How to measure and enhance LLM agents' risk awareness?*

To address these questions, we propose a judgment mechanism with information-theoretic optimization to probe safety boundaries and enhance agents' risk self-awareness through dual-objective training. Specifically, our judgment mechanism quantifies internal risk awareness of LLMs by integrating harmfulness assessment to measure the potential risk of agent responses with consistency evaluation using semantic similarity metrics across multiple responses. By applying Minimum Description Length (MDL) principles from information theory [19, 20, 21], which automatically identifies natural data breakpoints without manual parameter tuning or overfitting to noise, we optimize thresholds that partition agent behaviors into consistent safety, uncertainty, and consistent unsafety, offering a principled and interpretable approach to reveal safety boundaries. Guided by these boundaries, a dual-objective training strategy is developed to enhance consistent safety behaviors through supervised fine-tuning (SFT) and reduce harmful outputs via unlearning techniques. In the experiments, we explore LLM agent risk awareness using our judgment mechanism to identify safety boundaries. Subsequently, evaluation on multiple benchmarks demonstrates substantial improvements in risk awareness while preserving task execution functionality. The primary contributions of this work are: i) A judgment mechanism integrating harmfulness and consistency across multiple interactions to evaluate agent risk awareness beyond binary classifications, ii) MDL-based boundary optimization automatically identifying natural safety thresholds without manual tuning, and iii) A dual-objective training strategy combining SFT and unlearning techniques toward consistent safety behaviors.

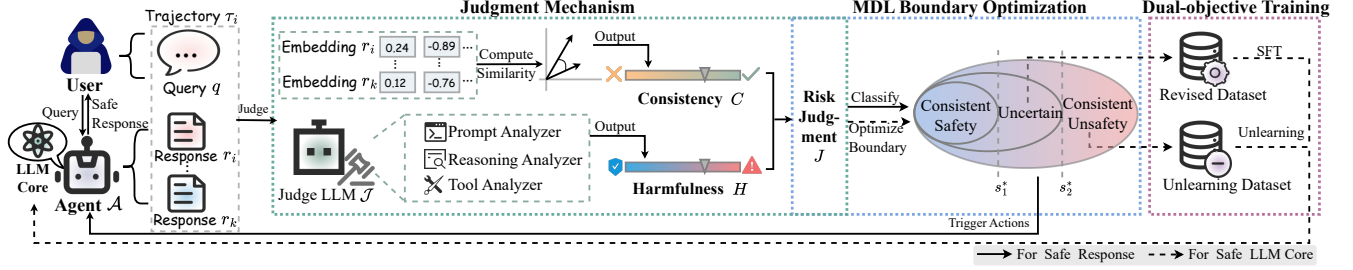


Fig. 1: Our framework introduces a judgment mechanism that enables boundary optimization, which then guides dual-objective training.

2. METHODOLOGY

Current safety approaches for LLM agents rely on binary safe or unsafe classifications [11, 10], failing to capture behavioral consistency across interactions or assess whether defenses enhance internal risk awareness rather than merely filtering outputs, thus limiting deployment in complex scenarios. We probe safety boundaries using a judgment mechanism integrating harmfulness and consistency assessment, followed by dual-objective training for consistent safe behaviors, as illustrated in Fig. 1.

2.1. Judgment Mechanism

To enable a comprehensive evaluation, multiple responses are generated from the target LLM agent $\mathcal{A}$ to the same input using temperature sampling to introduce natural variation. Given an input query q , a set of k trajectories $\mathcal{T} = \{\tau_1, \tau_2, \dots, \tau_k\}$ is generated where each trajectory τ_i represents the agent's complete response r_i containing user prompts, reasoning steps, tool interactions, and final outputs.

For harmfulness assessment, we employ an LLM-as-a-Judge approach to evaluate trajectories of the target LLM agent $\mathcal{A}$. The judge LLM $\mathcal{J}$ performs structured analysis of command patterns, prompt characteristics, and potential safety violations. For each trajectory $\tau_i \in \mathcal{T}$, the harmfulness function is defined as: $H(\tau_i) = \mathcal{J}(\tau_i) \in [0, 1]$, where $\mathcal{J}(\tau_i) \in [0, 1]$ represents the score given by the judge LLM $\mathcal{J}$, indicating the degree of harmfulness from none to maximum. To assess the overall harmfulness of multiple trajectories, we aggregate the harmfulness scores, so the final score H is computed as: $H(\mathcal{T}) = \frac{1}{k} \sum_{i=1}^k H(\tau_i)$, which represents the average harmfulness score of all generated trajectories of the target LLM agent $\mathcal{A}$. For consistency evaluation, we compute semantic similarity across the generated agent responses $\{r_1, r_2, \dots, r_k\}$ to capture consistency in the safety reasoning and final decision-making of the target LLM agent $\mathcal{A}$. The consistency score is calculated as:

$$C(\mathcal{T}) = \frac{1}{\binom{k}{2}} \sum_{1 \leq i_1 < i_2 \leq k} \cos(\phi(r_{i_1}), \phi(r_{i_2})), \quad (1)$$

where $\phi(\cdot)$ represents the sentence embedding function mapping responses to d -dimensional vectors in $\mathbb{R}^d$, and $\cos(\cdot, \cdot)$ computes cosine similarity defined as $\cos(u, v) = \frac{u \cdot v}{\|u\|_2 \|v\|_2}$. Inconsistent agent responses indicate high entropy in the agent's decision space, suggesting epistemic uncertainty in risk assessment. The risk judgment integrates harmfulness and consistency for the target LLM agent $\mathcal{A}$:

$$J(q, \mathcal{T}) = H(\mathcal{T}) + \alpha \cdot (1 - C(\mathcal{T})) \cdot (1 - H(\mathcal{T})), \quad (2)$$

where $\alpha \in [0, 1]$ controls the weight of uncertainty-induced risk. This judgment captures both direct harmfulness and epistemic un-

certainty. When responses are clearly harmful, inconsistency contributes minimal additional risk, whereas inconsistency in moderate or seemingly safe responses exposes uncertainty as a risk factor.

2.2. Boundary Optimization

Following the risk judgment process, establishing safety boundaries for target LLM agents becomes essential to capture distinct patterns in agent safety reasoning. Manual threshold selection relies on subjective choices and lacks principled justification. Hence, we propose an information-theoretic approach based on MDL principles to automatically identify these boundaries. MDL reframes boundary detection as a data compression problem, finding the model¹ that minimizes total description length. MDL determines optimal thresholds by finding classification schemes that minimize information needed to encode threshold locations and category assignments, automatically identifying natural breakpoints by balancing the cost of encoding the model structure against the cost of encoding the data.

Let $\mathcal{S} = \{J_1, J_2, \dots, J_n\}$ represent risk judgment scores computed using Eq. (2) for n different interactions. We establish thresholds $0 \leq s_1 < s_2 \leq 1$ to partition scores into $\mathcal{R}_{c+}$ for consistent safety ($J < s_1$), $\mathcal{R}_{u^o}$ for uncertainty ($s_1 \leq J < s_2$), and $\mathcal{R}_{c-}$ for consistent unsafety ($J \geq s_2$). Let n_ℓ denote the number of samples in each region, with corresponding empirical probabilities $p_\ell = \frac{n_\ell}{n}$ for $\ell \in \{c^+, u^o, c^-\}$. According to the MDL principle, the total description length for encoding both the model and the data is:

$$\mathcal{L}(s_1, s_2) = \mathcal{L}_{\text{model}}(s_1, s_2) + \mathcal{L}_{\text{data}}(s_1, s_2). \quad (3)$$

Here, "model" is the classification scheme defined by the threshold pair (s_1, s_2) . The model complexity $\mathcal{L}_{\text{model}}(s_1, s_2)$ represents the information required to specify this scheme, including $\log_2(3)$ bits for encoding the three-category structure and $2 \log_2(\frac{1}{\Delta})$ bits for encoding the two threshold parameters, where Δ represents the discretization step size [22, 23], giving $\mathcal{L}_{\text{model}}(s_1, s_2) = \log_2(3) + 2 \log_2(\frac{1}{\Delta})$. The data encoding cost $\mathcal{L}_{\text{data}}(s_1, s_2)$ represents the information needed to encode category assignments for each risk judgment score. According to Shannon's theorem [22], the optimal encoding length for an event with probability p is $-\log_2(p)$ bits. The total data encoding cost is

$$\mathcal{L}_{\text{data}}(s_1, s_2) = - \sum_{\ell \in \{c^+, u^o, c^-\}} n_\ell \log_2(p_\ell + \epsilon), \quad (4)$$

where $\epsilon = 10^{-10}$ prevents numerical instability for empty categories. Well-placed boundaries create homogeneous categories, reducing data encoding cost. To find such boundaries, the optimal

¹"Model" refers to any program used to describe and encode data patterns, including statistical models and algorithms.

threshold pair (s_1^*, s_2^*) is determined by solving:

$$(s_1^*, s_2^*) = \arg \min_{s_1, s_2} \mathcal{L}(s_1, s_2), \quad (5)$$

subject to $0 \leq s_1 < s_2 \leq 1, n_\ell \geq n_{\min},$

where $n_{\min}$ ensures statistical significance per category. The optimization can update thresholds incrementally as new risk judgments arrive, enabling boundary tracking of evolving agent behavior patterns. With optimal thresholds established, the classification captures distinct behavioral patterns in agent safety reasoning: *Consistent Safety* ($J < s_1^*$): Agent $\mathcal{A}$ consistently recognizes risks and provides appropriate safety responses; *Uncertainty* ($s_1^* \leq J < s_2^*$): Agent $\mathcal{A}$ inconsistently assesses risks, creating unpredictable behaviors; and *Consistent Unsafety* ($J \geq s_2^*$): Agent $\mathcal{A}$ consistently fails to identify harmful content, producing dangerous outputs. The classification triggers different actions to deliver safe responses to user queries: direct output for Consistent Safety and corrective safe responses for both Uncertainty and Consistent Unsafety.

2.3. Dual-objective Training Strategy

After MDL-based boundary optimization provides classification thresholds to categorize agent responses, we introduce a dual-objective training strategy that targets specific regions to enhance safety awareness and response consistency.

SFT for Consistency Enhancement. To improve response consistency for LLM agent $\mathcal{A}$, we address responses in the *Uncertainty* by employing SFT to teach the agent consistent safety patterns. When a query q generates k trajectory samples $\{\tau_i\}_{i=1}^k$ with risk judgment $J(q, \{\tau_i\}) \in [s_1^*, s_2^*]$, judge LLM $\mathcal{J}$ analyzes inconsistent responses and produces a corrected trajectory τ^s demonstrating safe reasoning [24]. The revised dataset is constructed as $\mathcal{D}_C = \{(q_j, \tau_j^s) : J(q_j, \{\tau_i^{(j)}\}_{i=1}^k) \in [s_1^*, s_2^*]\}_{j=1}^{N_c}$, where q_j represents the j -th query that falls into the *Uncertainty* and N_c is the total number. Each pair (q_j, τ_j^s) connects queries that cause uncertain responses with their corrected safe responses, turning the judge’s safety knowledge into training examples for the agent. The SFT objective minimizes negative log-likelihood loss to optimize agent parameters toward producing consistent responses:

$$\mathcal{L}_{\text{SFT}} = -\mathbb{E}_{(q, \tau^s) \sim \mathcal{D}_C} \left[\sum_{t=1}^{|\tau^s|} \log P_\theta(\tau_t^s | q, \tau_{<t}^s) \right], \quad (6)$$

where τ_t^s and $\tau_{<t}^s$ represent the current and preceding tokens respectively, enabling the agent to learn consistent safety patterns.

Unlearning for Unsafe Elimination. To decrease harmful response generation, we address *Consistent Unsafety* where agents consistently elicit harmful answers across multiple interactions, indicating deeply embedded unsafe patterns requiring unlearning. We employ gradient ascent unlearning to reduce the probability of generating consistently unsafe behaviors. When agent $\mathcal{A}$ generates trajectories $\{\tau_i\}_{i=1}^k$ for query q with risk judgment $J(q, \{\tau_i\}) \geq s_2^*$, these cases are collected for unlearning. The unlearning dataset is constructed as: $\mathcal{D}_H = \{(q_j, \tau_j^h) : J(q_j, \{\tau_i^{(j)}\}_{i=1}^k) \geq s_2^*\}_{j=1}^{N_h}$, where q_j represents the j -th query whose responses fall into the unsafe category, τ_j^h denotes one representative unsafe trajectory, and N_h is the total number of unsafe cases. The unlearning objective reduces the generation probability of harmful responses by minimizing their log-likelihood:

$$\mathcal{L}_{\text{unlearn}} = \mathbb{E}_{(q, \tau^h) \sim \mathcal{D}_H} \left[\sum_{t=1}^{|\tau^h|} \log P_\theta(\tau_t^h | q, \tau_{<t}^h) \right], \quad (7)$$

Table 1: Safety performance on InjecAgent and AgentHarm

Model	Method	InjecAgent		AgentHarm	
		DH-ASR	DS-ASR	H-Score	ASR
Qwen	Vanilla	0.3418	0.5076	0.2256	0.5582
	SafeMerge	0.2440	0.1777	0.2059	0.3652
	TaskShield	0.3050	0.4697	0.3348	0.5556
	Ours	0.0239	0.0338	0.1452	0.2083
	Ours ⁺	0.0303	0.0510	0.1763	0.2667
LLaMA	Vanilla	0.4121	0.6649	0.4370	0.7995
	SafeMerge	0.1410	0.2005	0.2826	0.5882
	TaskShield	0.2487	0.4134	0.3973	0.6156
	Ours	0.0447	0.1113	0.0016	0.0294
	Ours ⁺	0.1355	0.1217	0.0027	0.0172

Table 2: Task functionality on R-Judge and AgentHarm

Model	Method	R-Judge		AgentHarm	
		Accuracy	Precision	B-Score	PNA
Qwen	Vanilla	0.7360	0.7231	0.3567	0.3727
	Ours ⁺	0.8408	0.9746	0.4309	0.4324
LLaMA	Vanilla	0.8467	0.8374	0.3602	0.3472
	Ours ⁺	0.8291	0.9839	0.5569	0.4615

where τ_t^h and $\tau_{<t}^h$ represent current and preceding tokens respectively, enabling the agent to forget dangerous response patterns.

Overall, the dual-objective training framework integrates consistency enhancement and unsafe behavior elimination:

$$\theta^* = \arg \min_{\theta} [\mathcal{L}_{\text{SFT}}(\theta; \mathcal{D}_C) + \mathcal{L}_{\text{unlearn}}(\theta; \mathcal{D}_H)], \quad (8)$$

ensuring the agent develops stable safety reasoning while removing harmful behaviors, achieving improved consistency and enhanced safety while preserving general capabilities.

3. EXPERIMENTS

3.1. Setup

We present the experimental configuration across models, datasets, baselines, and metrics.

Models, Datasets and Tasks. We employ two large language models as the core components of our LLM agents: Qwen2.5-7B-Instruct and LLaMA-3.1-8B. For the judge LLM, we use GPT-4 to perform judgment. Our experimental evaluation is conducted on three benchmarks. InjecAgent [11] contains two attack categories: Direct Harm (DH) and Data Stealing (DS), with 1,054 test cases generated from 17 user scenarios and 62 attack scenarios. AgentHarm [10] contains 110 agent behaviors across 11 harm categories, each with harmful and benign counterparts. R-Judge [25] comprises 569 records of multi-turn agent interaction, encompassing 27 key risk scenarios among 5 application categories and 10 risk types.

Baseline. We compare against several baseline approaches for LLM agent safety. Vanilla LLM Agent serves as the baseline without any defense mechanisms. SafeMerge [26] selectively merges fine-tuned and safety-aligned model layers based on cosine similarity to preserve safety alignment. TaskShield [12] verifies whether each instruction and tool call contributes to user-specified goals to

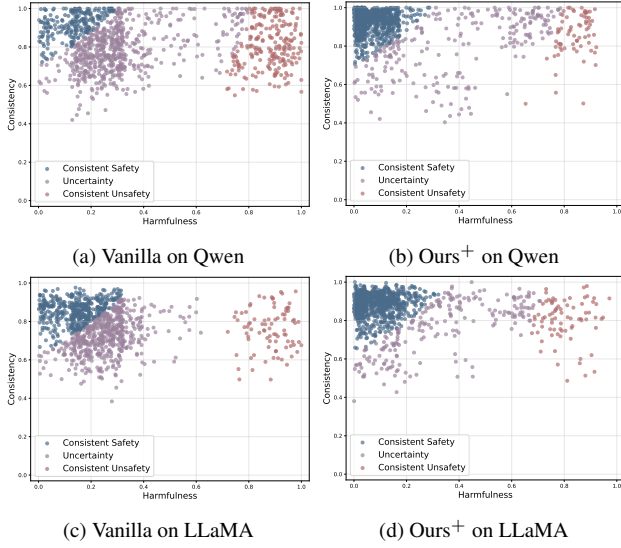


Fig. 2: Risk awareness on InjecAgent: Vanilla vs. Ours⁺ Agent.

defend against indirect prompt injection attacks. Additionally, our approach incorporates a judgment mechanism to assess safety risks during agent execution, while Ours⁺ applies the dual-objective training strategy to fine-tune the LLM with safety-aligned data.

Evaluation Metrics. We assess safety and task performance aspects using multiple metrics. For safety evaluation, Attack Success Rate (ASR) measures the percentage of successful attacks, with lower values indicating better security, and Harmful Score (H-Score) quantifies harmful behaviors when exposed to unsafe datasets. For task performance, Performance Under No Attack (PNA) evaluates task completion ability under normal conditions, Benign Score (B-Score) measures performance on safe tasks, and Accuracy and Precision evaluate response correctness and quality respectively.

3.2. Evaluation

We validate our approach across three key dimensions: safety performance, task functionality, and risk awareness.

Safety Performance. We evaluate safety performance of different methods across InjecAgent and AgentHarm benchmarks on various LLM core models, shown in Table 1. On InjecAgent, SafeMerge shows moderate ASR reductions compared to vanilla agents across both attack scenarios on Qwen and LLaMA, but provides insufficient protection. In contrast, TaskShield exhibits limited safety performance with minimal ASR decreases. Our approach consistently outperforms all baseline methods on both Qwen and LLaMA, achieving the lowest ASR values and highlighting the importance of integrated harmfulness and consistency assessment in our judgment mechanism. Ours⁺, which applies the dual-objective training strategy to improve the LLM core without executing the judgment mechanism, shows slightly higher ASR compared to our approach but still maintains robust safety performance across diverse threat scenarios. On AgentHarm, both our approach and Ours⁺ achieve lower H-Score and ASR than other methods, particularly on LLaMA, confirming the effectiveness of our judgment mechanism and dual-objective training in eliminating unsafe behaviors.

Task Functionality. We evaluate task functionality to ensure safety improvements do not compromise task execution capabilities, as shown in Table 2. On R-Judge, when using Qwen as the core

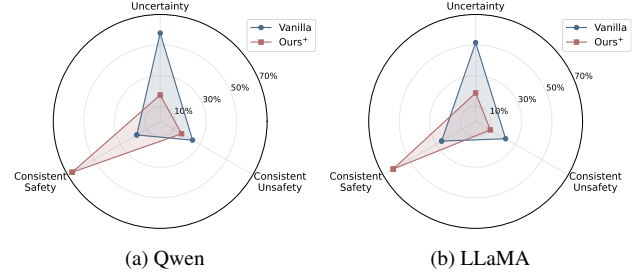


Fig. 3: Risk awareness on AgentHarm: Vanilla vs. Ours⁺ Agent.

model, Ours⁺ demonstrates improvements in both response correctness and quality compared to the vanilla agent. For LLaMA, accuracy slightly decreases while precision increases, suggesting improved reliability in risk detection despite a minor reduction in overall correctness. On AgentHarm, Ours⁺ shows improved B-Score and PNA across both Qwen and LLaMA compared to vanilla agents, with improvements particularly evident on LLaMA, demonstrating that safety-aligned training not only reduces harmful outputs but also improves the overall quality and reliability of agent responses.

Risk Awareness. To demonstrate how our approach transforms agent behaviors, we provide a visual analysis of risk awareness patterns for LLM agents, comparing vanilla and Ours⁺ across different benchmarks through MDL-based boundary optimization. Fig. 2 shows harmfulness and consistency distributions on InjecAgent. On Qwen, vanilla agents exhibit widespread distribution across all regions, with considerable presence in uncertainty and consistent unsafety zones, indicating poor risk awareness. In contrast, Ours⁺ concentrates most samples in the consistent safety region, with clear clustering in the low-harmfulness, high-consistency area. On LLaMA, a similar transformation occurs, with samples migrating from high-risk zones to the consistent safety region, demonstrating improved risk awareness and reliability. Fig. 3 illustrates the distribution changes on AgentHarm. On both Qwen and LLaMA, vanilla agents show significant presence in uncertainty and consistent unsafety regions. With Ours⁺, the distributions shift dramatically toward consistent safety, with notable reductions in uncertainty and unsafe classifications. Our dual-objective training enhances agents' internal risk perception, enabling more consistent safety behaviors.

4. CONCLUSION

In summary, we explore LLM agent risk awareness through a judgment mechanism that integrates harmfulness and consistency assessment through multiple interactions. Building on these measurements, we apply MDL-based boundary optimization to establish principled thresholds without manual parameter tuning that classify agent behaviors into consistent safety, uncertainty, or consistent unsafety, which then guide our dual-objective training strategy that combines supervised fine-tuning with unlearning techniques to effectively enhance risk awareness in LLM agents. Experimental results demonstrate that our approach enhances safety awareness without compromising task performance, with LLM agent behaviors shifting toward consistent safety. However, our approach has limitations as we rely on GPT-4 as the judge LLM for harmfulness assessment without exploring other models, and consistency evaluation incurs computational overhead, which will be addressed through future work exploring diverse judge models and developing more efficient evaluation methods.

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